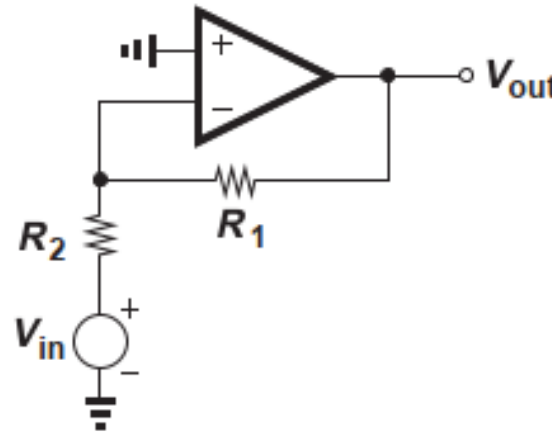
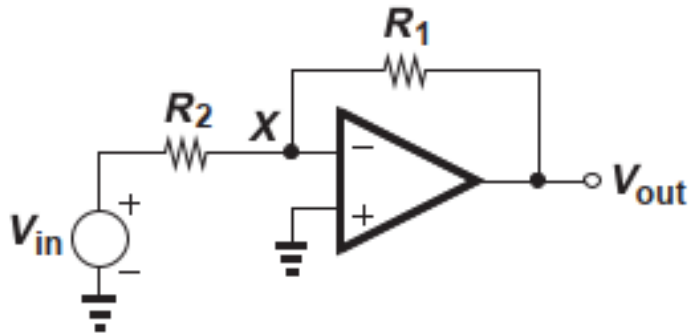


Operational Amplifiers: Inverting Amplifier



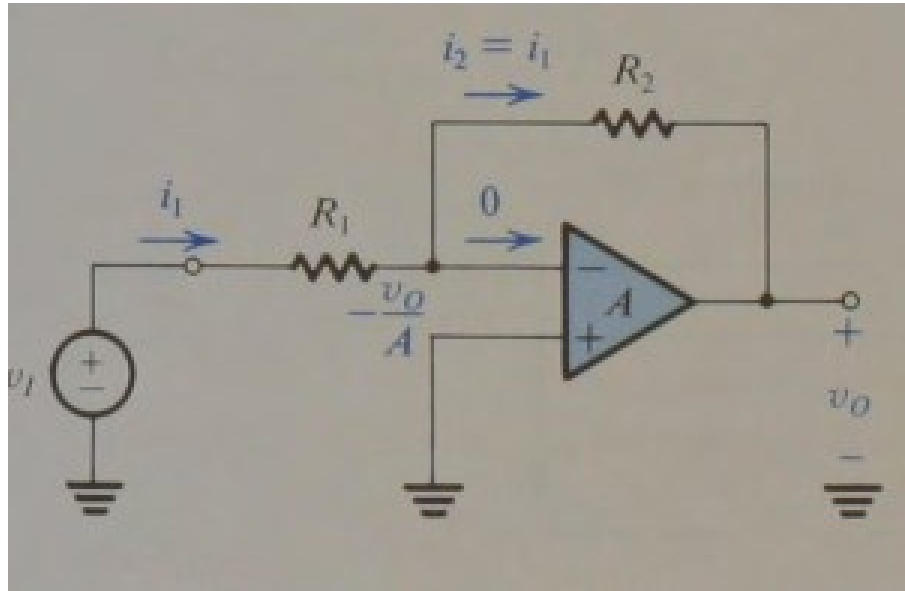
$$V_{out} = A_0(V_{in1} - V_{in2})$$
$$= -A_0 V_X.$$

$$\frac{V_{out}}{V_{in}} = -\frac{1}{\frac{1}{A_0} + \frac{R_2}{R_1} \left(\frac{1}{A_0} + 1 \right)}$$
$$= -\frac{1}{\frac{R_2}{R_1} + \frac{1}{A_0} \left(1 + \frac{R_2}{R_1} \right)}.$$

Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

$$\frac{V_{out}}{V_{in}} \approx -\frac{R_1}{R_2} \left[1 - \frac{1}{A_0} \left(1 + \frac{R_1}{R_2} \right) \right]$$

Operational Amplifiers: Inverting Amplifier

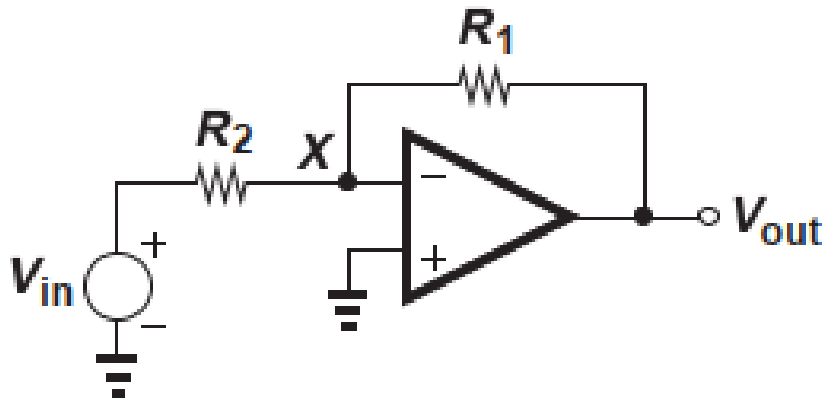


Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

$$i_1 = \frac{v_I - (-v_O/A)}{R_1} = \frac{v_I + v_O/A}{R_1}$$

$$\begin{aligned} v_O &= -\frac{v_O}{A} - i_1 R_2 \\ &= -\frac{v_O}{A} - \left(\frac{v_I + v_O/A}{R_1} \right) R_2 \end{aligned}$$

$$\frac{v_O}{v_I} = \frac{-R_2/R_1}{1 + (1 + R_2/R_1)/A}$$



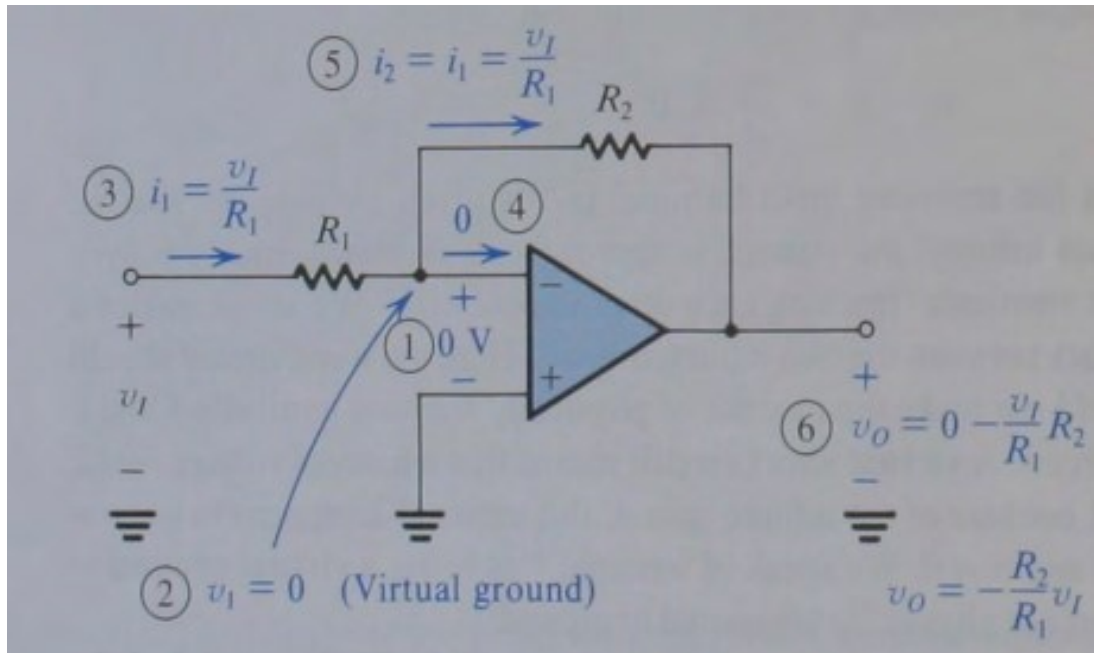
Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

- For an infinite gain:

$$\frac{0 - V_{out}}{R_1} = \frac{V_{in}}{R_2}$$

$$\frac{V_{out}}{V_{in}} = \frac{-R_1}{R_2}$$

Operational Amplifiers: Inverting Amplifier



- For an infinite gain:

$$v_2 - v_1 = \frac{v_O}{A} = 0$$

$$i_1 = \frac{v_I - v_1}{R_1} = \frac{v_I - 0}{R_1} = \frac{v_I}{R_1}$$

$$\begin{aligned} v_O &= v_1 - i_1 R_2 \\ &= 0 - \frac{v_I}{R_1} R_2 \end{aligned}$$

$$\frac{v_O}{v_I} = -\frac{R_2}{R_1}$$

Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

OpAmps: Inverting and Non-Inverting Amplifier

